

The recombination mechanism and defect energy levels of Ag-doped $\text{ZnSnN}_2/\text{ZnSnN}_2$ pn homojunction solar cells

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Fan Ye, , Qian Gao, Qiao Tang, , Yi-Zhu Xie, , Ying Liu, Chun-Yan Yu, Fan Wang, , Dong-Ping Zhang, and Xing-Min Cai

AFFILIATIONS

Key Laboratory of Optoelectronic Devices and Systems of Ministry of Education and Guangdong Province, Shenzhen Key Laboratory of Advanced Thin Films and Applications, College of Physics and Optoelectronic Engineering, and State Key Laboratory of Radio Frequency Heterogeneous Integration, Shenzhen University, Shenzhen, Guangdong 518060, China

^{a)} Author to whom correspondence should be addressed: caixm@szu.edu.cn

ABSTRACT

The photovoltaic performance of ZnSnN_2 homojunction has never been studied. Here, ZnSnN_2 homojunction solar cells with Ag-doped ZnSnN_2 as the *p*-type layer are fabricated with magnetron sputtering deposition, and the mechanism limiting the photovoltaic performance is revealed. The junction is abrupt. The current transport is dominated by thermionic emission below 320 K and generation-recombination (GR) in the space charge region in 330–380 K. Below 320 K, the short circuit current density and the open circuit voltage are limited by non-radiative interface recombination. The open circuit voltage loss also results from interfacial barrier height inhomogeneity, which obeys the Gaussian distribution model or the temperature fluctuation of interface barrier heights for holes. At higher temperatures, deep energy levels in the middle of the bandgap result in the GR in the space charge region. One shallow energy level at 0.13 eV and one interface energy level at 0.72 eV above the valence band maximum are observed. The former is likely induced by substitutional Ag-doping.

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Thin film solar cells have significantly advanced the development of solar energy technologies. To date, numerous promising candidate materials have emerged, including $\text{Cu}_2\text{ZnSnS}_4$, Sb_2Se_3 , and ZnSnN_2 (ZTN).^{1–5} ZTN is a ternary nitride semiconductor and is attracting attention due to its tunable direct bandgap and high absorption coefficient within the solar spectrum, non-toxicity, and Earth-abundance of the constituent elements, demonstrating great potential for photovoltaic, photocatalytic, and light-emitting applications.^{1,2,6–11}

In early studies, ZTN is found to exhibit degenerate electron concentrations, mainly due to the low formation energies of donor defects, including Sn_{Zn} (Sn on Zn sites) and unintentional O_{N} (O on N sites), compared to acceptor defects like Zn_{Sn} (Zn on Sn sites).^{6–8} Subsequent methods, including non-stoichiometric growth with higher Zn content and hydrogen doping or passivation, are effective in reducing the electron concentration to the magnitude of 10^{16} cm^{-3} .^{9,12–16} Various ZTN-based devices, such as Schottky and heterojunction solar cells and photodetectors, have been fabricated.^{17–22} More recently, its *p*-type conduction has been achieved with doping Li, Ba, Al, Ga, or Ag^{3,23–25} and rectifying ZTN pn

homojunctions³ with Al- or Ga-doping and no photovoltaic effect have been constructed.

However, ZTN homojunctions with the photovoltaic effect have never been reported. In this work, we demonstrate that ZTN homojunction solar cells with Ag-doped ZnSnN_2 (AZTN) as the *p*-type layer can be prepared using magnetron sputtering, which is compatible with industry production. The properties of the solar cells are studied, and the mechanism that limits the open circuit voltage (V_{oc}) and short circuit current density (J_{sc}) is revealed.

To obtain the properties of thin films, K9 glass substrates and Si are used as the substrates. Indium tin oxide (ITO) coated glass is used as the substrate to prepare the homojunctions with the structure of glass/ITO/ZTN/AZTN/Pt. The details about preparing and characterizing the films, including ZTN, AZTN, and Pt and the homojunctions, are in the [supplementary material](#) (the steps to prepare the homojunctions are in Fig. S1).

The XRD patterns of the ZTN and AZTN thin films are in Fig. S2, [supplementary material](#). ZTN is wurtzite^{7,9} while AZTN is amorphous or nanocrystalline.²⁶ The surface morphology of the thin films

is in Fig. S3. The reflectance (R_c), transmittance (T_r), and Tauc plots are in Fig. S4. The film thickness d for ZTN and AZTN is 334.3 and 66.6 nm, respectively. The optical bandgap E_g for ZTN and AZTN is 2.02 and 1.52 eV,^{3,23,24} respectively. The electrical properties of the films are characterized using Hall effect measurements. The ZTN layer is n -type conductive with an electron density of $1.51 \times 10^{17} \text{ cm}^{-3}$ while the AZTN layer is p -type conductive with a hole density of $1.59 \times 10^{15} \text{ cm}^{-3}$ and a mobility μ_p of $2.68 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$.

Figure 1(a) shows the dark current density (J) vs the applied voltage (V) curve of the homojunction (glass/ITO/ZTN/AZTN/Pt). The rectification suggests the formation of a p - n junction at the ZTN/AZTN interface since ITO and Pt form Ohmic contacts with the adjacent ZTN and AZTN layers.^{25,26} Under illumination, the homojunction exhibits a weak but discernible photovoltaic response [Fig. 1(b)], with a photoelectric conversion efficiency (E_{ff}) of 0.0003%, along with a V_{oc} of 0.21 V, a J_{sc} of 0.0046 mA/cm^2 , and a fill factor (FF) of 31.24%.

Figure 1(c) shows the dark forward double-logarithmic J - V curve, which can be divided into three distinct regions:²⁷ an Ohmic region (slope ≈ 1) at low bias voltages, a trap-filled-limited (TFL) region (slope > 2) at intermediate voltages, where the current increases sharply due to the filling of trap states, and a child's law region (slope ≈ 2) at higher voltages, where space charge limited current (SCLC) dominates. The transition voltage from the Ohmic to the TFL region is defined as the trap-filled-limit voltage ($V_{TFL} = 0.52 \text{ V}$). The trap density (N_{trap}) can be estimated²⁸ since $N_{trap} = 2\epsilon_r\epsilon_0 V_{TFL} / (qd^2)$, where $\epsilon_r = 11^2$ is the relative dielectric constant of ZTN, ϵ_0 is the vacuum permittivity, q is the elementary charge, and d is the thickness of the AZTN layer. The very possible origin of these traps is deep defects in the bulk. The calculated value of N_{trap} for the

homojunction is $1.43 \times 10^{17} \text{ cm}^{-3}$, larger than that (10^{14} cm^{-3})¹⁹ of $\text{Cu}_2\text{ZnSnS}_4/\text{ZTN}$, resulting in low efficiency.

The capacitance-voltage (C - V) curve measured at 10 kHz is shown in Fig. 1(d). In general, the measured capacitance C of a homojunction consists of the space charge capacitance (C_{SC}) and the interface state capacitance (C_{IS}) in parallel. Therefore, $C = C_{IS} + C_{SC}$.²⁹ C_{SC} equals³⁰ $A_{SC}/(V_{bi} - V)^{1/2}$ and $A_{SC} = \{0.5\epsilon_r\epsilon_0 N_A N_D / (N_A + N_D)\}^{1/2}$, where N_A and N_D are the doping concentrations in the p -type and n -type layers, and V_{bi} is the built-in potential. Using C_{IS} , A_{SC} , and V_{bi} as the fitting parameters, the C - V data can be fitted to $C = C_{IS} + A_{SC}/(V_{bi} - V)^{1/2}$ with the least squares methods. The results show that $C_{IS} = 2.0 \times 10^{-7} \text{ F/cm}^2$, $A_{SC} = 9.1 \times 10^{-7} \text{ V}^{0.5} \text{ F/cm}^2$, and $V_{bi} = 0.66 \text{ V}$ [Fig. 1(e)]. N_A is estimated to be $1.07 \times 10^{17} \text{ cm}^{-3}$ since A_{SC} can be simplified to be $\{0.5\epsilon_r\epsilon_0 N_A\}^{1/2}$ according to the Hall effect measurement. Therefore, the p n junction is abrupt. In addition, since Hall effect measurements show that $N_D > N_A$, V_{bi} is related to the built-in potential of the holes, as shown in the energy band diagram in Fig. S5.^{21,25,31,32}

The temperature-dependent J - V curves of the homojunction are in Figs. S6(a) and S6(b). The current transport mechanism^{33,34} is thermionic emission (TE) in 190–320 K and generation recombination (GR) in the space charge region in 330–380 K [Figs. S6(c) and S6(d)]. The Cheungs' method³⁵ is used to extract the barrier height ϕ_b , ideality factor n , and series resistance R_s . The barrier height is inhomogeneous and obeys the Gaussian distribution model^{36–38} in which the relation between ϕ_b and temperature T is

$$\phi_b = \overline{\phi}_b - q\sigma_0^2/(2kT), \quad (1)$$

and that between n and T is

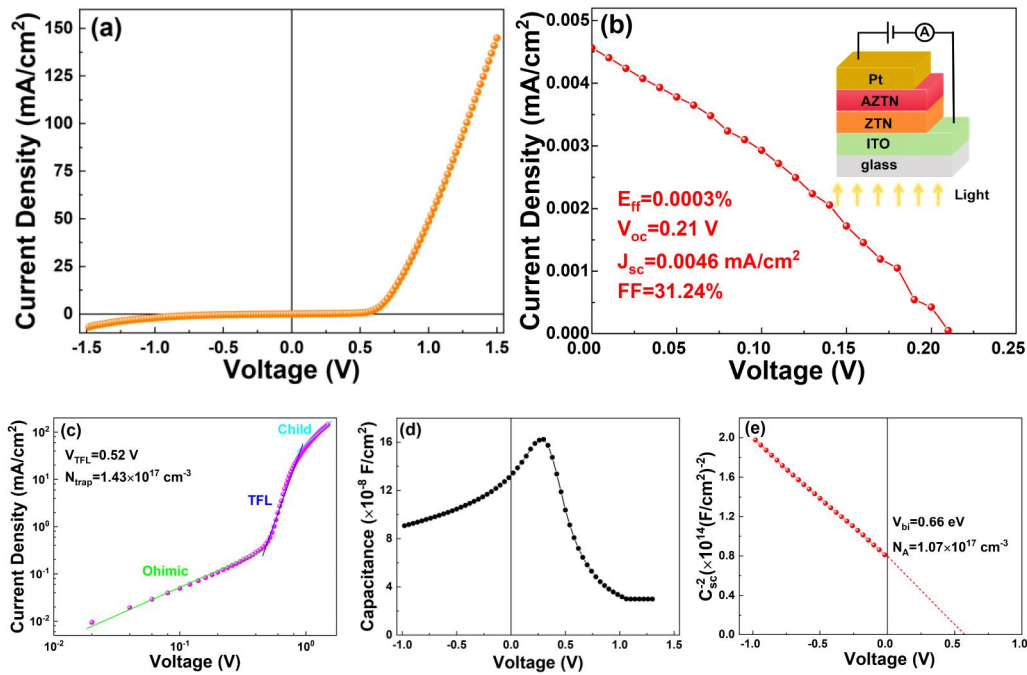


FIG. 1. The J - V curve of the homojunction measured in darkness (a) and under illumination (b). (c) Double-logarithmic J - V plot; (d) The C - V curve. (e) The C_{SC}^{-2} - V curve.

$$n^{-1} - 1 = -\rho_2 + q\rho_3/(2kT), \quad (2)$$

where k is the Boltzmann constant, $\overline{\phi_b}$ and σ_0 are the standard barrier height and deviation at zero-bias while ρ_2 and ρ_3 are two coefficients (Fig. S7). The effective Richardson constant (A^{**}) obtained from the modified Richardson plot is **14.58 A cm⁻² K⁻², near the theoretical¹⁴ 14.40 A cm⁻² K⁻².**

The J-V curves of the solar cell can also be described by the single-diode equation,³⁹

$$J = J_0 \{ \exp [q(V - JR_s)/(nkT)] - 1 \} + GV - J_L, \quad (3)$$

where J_0 is the diode current density, G is the shunt conductance, and J_L is the photo-induced current density ($J_L = 0$ under darkness). The four parameters J_0 , R_s , n , and G can be extracted with the analytic method³⁹ and those from the JV curve at 300 K are in Fig. S8. All the extracted data are in Table S1. Compared with the Cu₂ZnSnS₄/ZTN heterojunction,¹⁹ the homojunction has higher R_s , indicating restricted carrier transport, possibly due to the relatively higher density of defect states. From the plot of $\ln J_0$ vs $1/(kT)$ (Fig. S9), the current is also controlled by thermionic emission below 320 K and above that by GR in the space charge region.

Figure 2(a) shows that the Richardson plot is not linear in 210–320 K. Applying Eq. (2) to the ideality factors n in this region yields $\rho_2 = -0.06$ and $\rho_3 = -0.03$ V [Fig. 2(b)]. In the single-diode model, J_0 is written as^{39–44}

$$J_0 = J_{00} \exp[-q\phi_b/(nkT)], \quad (4)$$

where J_{00} depends on the specific recombination mechanism. From Eq. (4), ϕ_b at each temperature can be calculated and listed as ϕ_{b1} in Table S1 since J_0 and n are known while J_{00} actually equals $A^{**}T^2$ in 210–320 K due to the fact that the dark forward current is dominated by TE in this range. The obtained ϕ_{b1} is much larger than those obtained from the Cheungs' method. In addition, applying Eq. (1) to

ϕ_{b1} [Fig. 2(c)] gives negative σ_0^2 , which is unphysical. From the modified Richardson plot based on the negative σ_0^2 [Fig. 2(d)], the extracted A^{**} is about 10^{-27} A cm⁻² K⁻², far smaller than the theoretical value.

If the ideality factor n is omitted from Eq. (4), J_0 in the single-diode model will be

$$J_0 = J_{00} \exp[-q\phi_b/(kT)]. \quad (5)$$

ϕ_b at each temperature can also be calculated and listed as ϕ_{b2} in Table S1. The obtained ϕ_{b2} is near to those obtained with the Cheungs' method. Applying Eq. (4) to ϕ_{b2} [Fig. 2(e)] yields $\sigma_0^2 = 0.026$ V² and $\overline{\phi_{b2}} = 1.13$ V. In Fig. 2(f), the modified Richardson plot based on $\sigma_0^2 = 0.026$ V² not only yields $\overline{\phi_{b2}} = 1.13$ V but also A^{**} of 13.46 A cm⁻² K⁻², which is near the theoretical value. Therefore, when the dark forward current is governed by TE in the single-diode model, the barrier height calculated from the diode current J_0 excluding the ideality factor n meets the Gaussian distribution model.

The n - T relation^{45,46} under nonradioactive interface recombination (IR) with tunneling enhancement is

$$n = E_{00}/(kT) \coth[E_{00}/(kT)], \quad (6)$$

where E_{00} is the characteristic tunneling energy while that under non-radiative bulk recombination (BR) with tunneling enhancement is

$$n^{-1} = \frac{1}{2} \left[1 - \frac{E_{00}^2}{3(kT)^2} + \frac{T}{T^*} \right], \quad (7)$$

where kT^* is the characteristic energy of the distribution of trap states near the band edge associated with band-tailing or potential fluctuation. Figure 3(a) shows the least squares method fitting of the ideality factors in Table S1 to Eq. (6) with $E_{00} = 58.70$ meV, and to Eq. (7) with $E_{00} = 27.40$ meV and $kT^* = 69.13$ meV. The top dashed line in Fig. 3(a) is from Eq. (7) with $E_{00} = 0$ meV and $kT^* = 69.13$ meV. Below 280 K, Eq. (6) gives a better fitting, indicating that below this

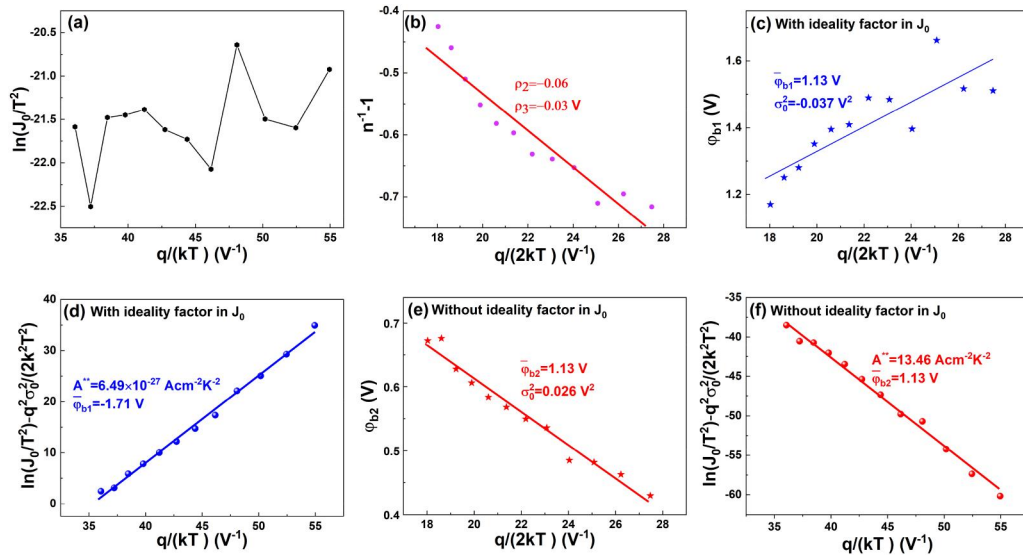


FIG. 2. (a) The curve of $\ln(J_0/T^2)$ vs $q/(kT)$. (b) The curve of $n^{-1} - 1$ vs $q/(2kT)$. The curve of ϕ_{b1} vs $q/(2kT)$ plot (c) and the modified Richardson plot (d) (both with n in J_0). The curve of ϕ_{b2} vs $q/(2kT)$ (e) and the modified Richardson plot (f) (both without n in J_0).

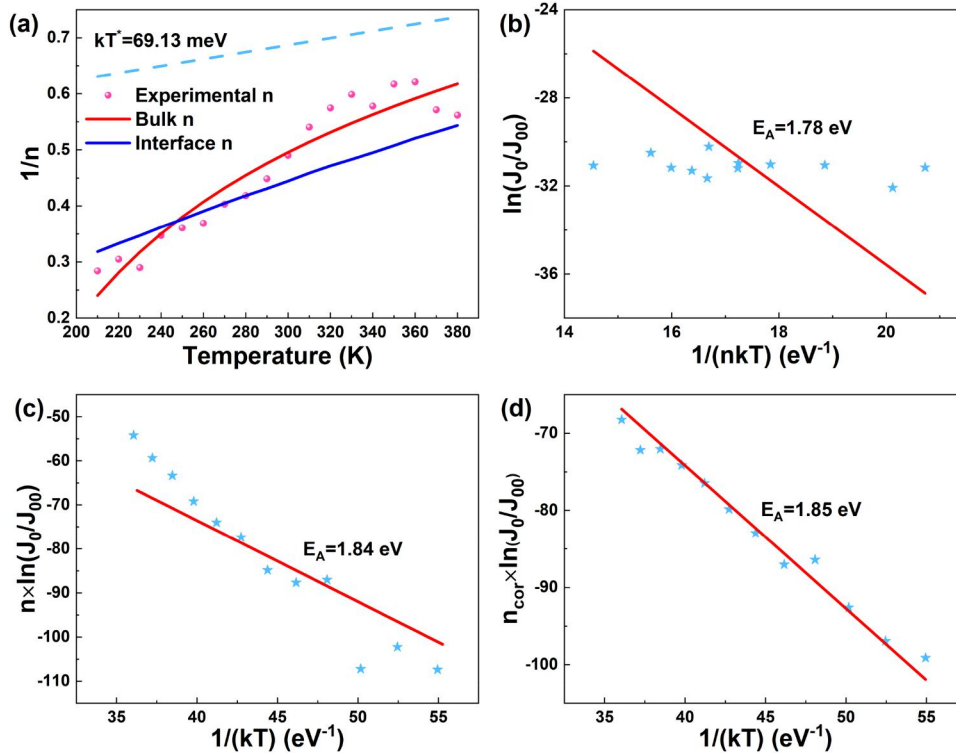


FIG. 3. (a) Fitting the ideality factor to Eqs. (6) and (7). (b) The curve of $\ln(J_0/J_{00})$ vs $1/(nkT)$. (c) The curve of $n \ln(J_0/J_{00})$ vs $1/(kT)$. (d) The curve of $n_{\text{cor}} \ln(J_0/J_{00})$ vs $1/(kT)$.

temperature, tunneling enhanced IR is dominant. The effective defect density N_{def} is calculated to be $1.32 \times 10^{19} \text{ cm}^{-3}$, from $E_{00} = hN_{\text{def}}^{0.5}/[2(m^* \epsilon_r \epsilon_0)^{0.5}]$, where h is the Plank constant and m^* is the effective electron mass (m^* is 0.12 times that of a rest electron^{47–50}). This high N_{def} which is from all the interface states that can provide the path for tunneling, is in agreement with the poor crystallization of ZTN and AZTN, leading to the relatively low J_{sc} .⁴⁶ The fitting to Eq. (7) is roughly acceptable in the whole range but better than Eq. (6) in the higher temperature range, indicating that tunneling enhanced BR gradually gets dominant at higher temperatures.

To obtain the activation energy E_A of the electrical transport below 320 K, Eq. (4) is rewritten as⁴⁶

$$J_0 = J_{00} \exp[-E_A/(nkT)]. \quad (8)$$

Figure 3(b) shows the curve of $\ln(J_0/J_{00})$ vs $1/(nkT)$, hardly any activation energy can be obtained while Fig. 3(c) shows the curve of $n \ln(J_0/J_{00})$ vs $1/(kT)$ with poor linearity observed. To solve these problems, the corrected ideality factor n_{cor} is calculated according to Eq. (6) with $E_{00} = 58.70 \text{ meV}$. Figure 3(d) shows the curve of $n_{\text{cor}} \ln(J_0/J_{00})$ vs $1/(kT)$, good linearity is observed. The obtained E_A is 1.85 eV, smaller than the bandgap of n -type ZTN, also implying IR dominance in this region.

Under IR, V_{oc} is controlled by the interfacial barrier height ϕ_H of the holes, and the theoretical V_{oc} at 300 K is 1.49 V (Fig. S5), much larger than the observed 0.21 V. V_{oc} controlled by IR and affected by barrier height inhomogeneity is^{39,42,44} $V_{\text{oc}} = n\bar{\phi}_b - qn\sigma_0^2/(2kT) - (nkT/q) \ln(qS_{\text{it}}N_V/J_L)$ according to Eqs. (3) and (5), where S_{it} is the IR

velocity. With the measured V_{oc} , J_L and the parameters obtained from the single-diode model, the calculated S_{it} is $1.77 \times 10^4 \text{ cm/s}$, near that of other semiconductors.⁴² In addition, V_{oc} according to Eqs. (3) and (8) is^{39,42,44} $V_{\text{oc}} = [\phi_{b1} - q\sigma^2/(2kT)] - (nkT/q) \ln(qS_{\text{it}}N_V/J_L)$, where $\sigma = 0.03 \text{ V}$ is the standard deviation of ϕ_{b1} (Fig. S10). From this, S_{it} at 300 K is calculated to be $1.36 \times 10^4 \text{ cm/s}$, with $q\phi_{b1} = 1.28 \text{ eV}$ at 300 K. Therefore, strong IR as well as barrier height inhomogeneity or the temperature fluctuation of interface barrier heights for holes results in low V_{oc} .

The current in 330–380 K is well fitted to the current formula due to GR in the space charge region with deep energy levels in the middle of the forbidden bandgap^{51,52} (Fig. S11). This suggests the GR in the space charge region is mainly due to deep energy levels in the middle of the forbidden bandgap.

The double-logarithmic J - V curves in 300–380 K under 1.0–1.5 V are in Fig. 4(a). These curves exhibit typical SCLC behavior, indicating that charge transport in the AZTN layer is dominated by a single trap level. In this regime, the current density is $J = 9\epsilon_r\epsilon_0\mu_p\theta V^2/(8d^3)$,⁵³ where θ is the trap factor. The values of θ at various temperatures can be extracted from the intercepts on the vertical axis in Fig. 4(a). The relation between the trap factor θ and the defect energy level E_t measured from the valence band maximum (E_v) is $\theta = (N_v/N_t) \exp[-E_t/(kT)]$, where N_v is the effective density of states in the valence band, and N_t is the concentration of traps at this energy level. By plotting $\ln\theta$ as a function of $1/(kT)$ [Fig. 4(b)], E_t is found to be 0.13 eV. Since the carrier density of the n -type ZTN is larger than that of p -type AZTN, E_t can be considered to be in the p side and in close proximity to E_v . This indicates it is a shallow acceptor level, most likely resulting from substituting Ag for Zn or Sn [Fig. 4(c)].

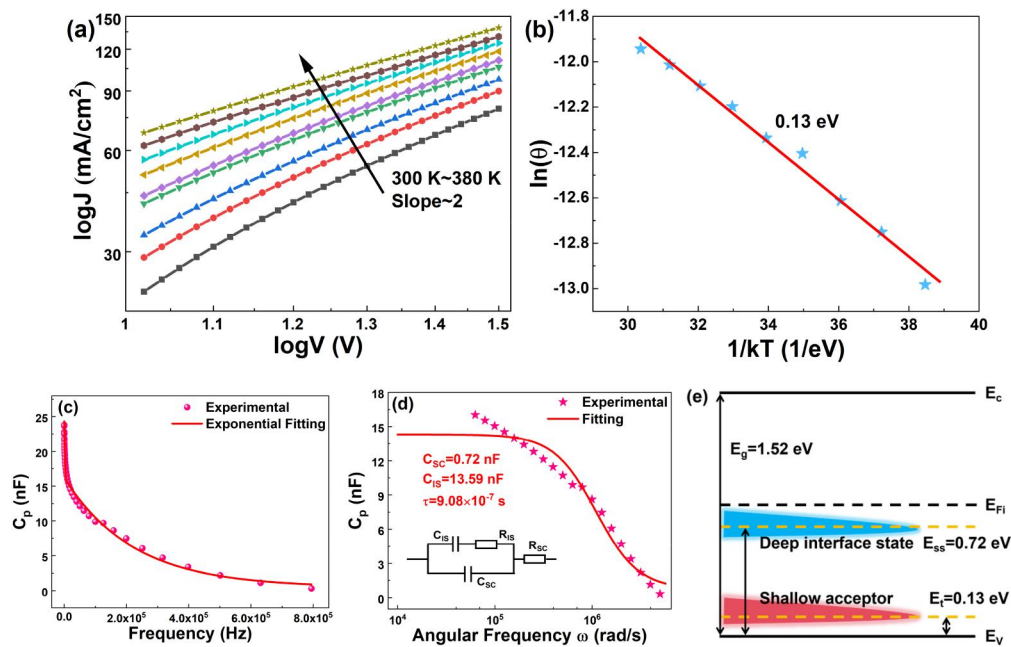


FIG. 4. (a) Double-logarithmic J–V curves in the SCLC regime in 300–380 K. (b) The curve of $\ln \theta$ vs $1/(kT)$. (c) The C_p -f curves and the exponential fitting. (d) The fitting to the C_p -f curve (inset: the equivalent circuit for fitting). (e) The energy levels.

Figure 4(c) shows the zero-bias admittance spectroscopy at 300 K. The capacitance (C_p) exhibits a strong frequency (f) dependence over the measured range. Fitting the C_p -f data with an exponential decay function [solid line in Fig. 4(c)] confirms that C_p decays exponentially with increasing frequency. This behavior is typically attributed to the dispersion effect of interface states: under high-frequency conditions, deep-level traps in quasi-equilibrium with the semiconductor Fermi level cannot respond sufficiently to the alternative current (AC) signal, resulting in a decrease in the measured capacitance.^{54–56} This frequency dependence can be described by an equivalent circuit model incorporating interface state response, in which⁵⁴ $C_p = C_{SC} + C_{IS}/[1 + (\omega\tau)^2]$, where $\omega = 2\pi f$ is the angular frequency and τ is the interface trap time constant of the interface states. The C_p -f data are fitted to this equation using the least squares method with C_{SC} , C_{IS} , and τ as the fitting parameters [Fig. 4(d)] and the fitting yields $C_{SC} = 0.72$ nF, $C_{IS} = 13.59$ nF, and $\tau = 9.08 \times 10^{-7}$ s. The interface state density $N_{ss} = C_{IS}/(qA_j)$ is calculated to be $5.31 \times 10^{11} \text{ eV}^{-1} \text{ cm}^{-2}$ ($A_j = 0.16 \text{ cm}^2$). The interface state energy level E_{ss} satisfies $E_{ss} - E_V = q(\phi - V)$, where $\phi = V_{bi} + kT/q \ln(N_V/N_A)$. Since the C_p -f measurement is performed at $V = 0$ V while $V_{bi} = 0.66$ V, substituting $N_V = 1.04 \times 10^{18} \text{ cm}^{-3}$ and $N_A = 1.07 \times 10^{17} \text{ cm}^{-3}$ yields $\phi = 0.72$ eV. Thus, $E_{ss} \approx E_V + 0.72$ eV, indicating that the interface state is located approximately 0.72 eV above E_V of the AZTN layer, corresponding to a deep-level defect [Fig. 4(e)].

The origin of this deep level is speculated based on the results in the literature. First, shallower isolated defects such as Sn_{Zn} (about 0.37 eV below the conduction band minimum E_c) and O_N ($E_c - 0.21$ eV) can be ruled out.⁶ Theoretical studies suggest that under Zn-rich and unintentional oxygen-containing conditions, the charge-neutral complex ($\text{Zn}_{\text{Sn}} + 2\text{O}_N$) is energetically dominant.^{6,57}

Nevertheless, in the cation-disordered interface environment formed during actual deposition, Zn_{Sn} and O_N may not pair ideally, potentially leading to acceptor-type complexes such as ($\text{Zn}_{\text{Sn}} + \text{O}_N$). Owing to incomplete charge compensation, such complexes may introduce deep levels near the midgap. In addition, the doped Ag atoms may also result in deep energy levels by forming different complexes with neighboring intrinsic vacancies or anti-site defects.

Figure 5(a) shows the depth profile of the charge density N_{CV} calculated from the C–V measurements and N_{DLCP} calculated from drive-level capacitance profiling (DLCP) measurements at 10 kHz. The variation of N_{CV} with depth X_{CV} ($X_{CV} = \epsilon_r \epsilon_0 A_j / C$) is given by $N_{CV} = C^3 (dC/dV)^{-1} / (q \epsilon_0 \epsilon_r A_j^2)$.¹⁹ The variation of N_{DLCP} with depth X_{DLCP} is given by⁵⁸ $N_{DLCP} = -C_0^3 / (2q \epsilon_r \epsilon_0 A_j^2 C_1)$, where C_0 and C_1 are the fitting parameters during the DLCP measurement while $X_{DLCP} = \epsilon_r \epsilon_0 A_j / C_0$. Since N_{CV} includes contributions from free carriers, bulk defects, and interface states, whereas N_{DLCP} results only from the former two, the interface state density N_i is $7.86 \times 10^{16} \text{ cm}^{-3}$ ($N_{CV} - N_{DLCP}$). The extracted N_i here is smaller than N_{def} from Fig. 3(a) since N_i here is just from those that can respond to the measurement frequency. It is higher as compared with other ZTN-based junctions,^{19,20} resulting in lower E_{ff} .

From the Nyquist plot of the homojunction [Fig. 5(b)], a single semicircular arc is observed. The inset illustrates the equivalent circuit used for fitting, which includes the series resistance R_s , the recombination resistance R_{rec} and the parallel capacitance C . The fitting yields $R_s = 44.77 \text{ } \Omega$, $R_{rec} = 132.50 \text{ } \Omega$, $C = 14.58$ nF, and shows good agreement with the experimental data. The recombination time constant (τ_{rec}) is about $1.93 \text{ } \mu\text{s}$, shorter than other ZTN-related solar

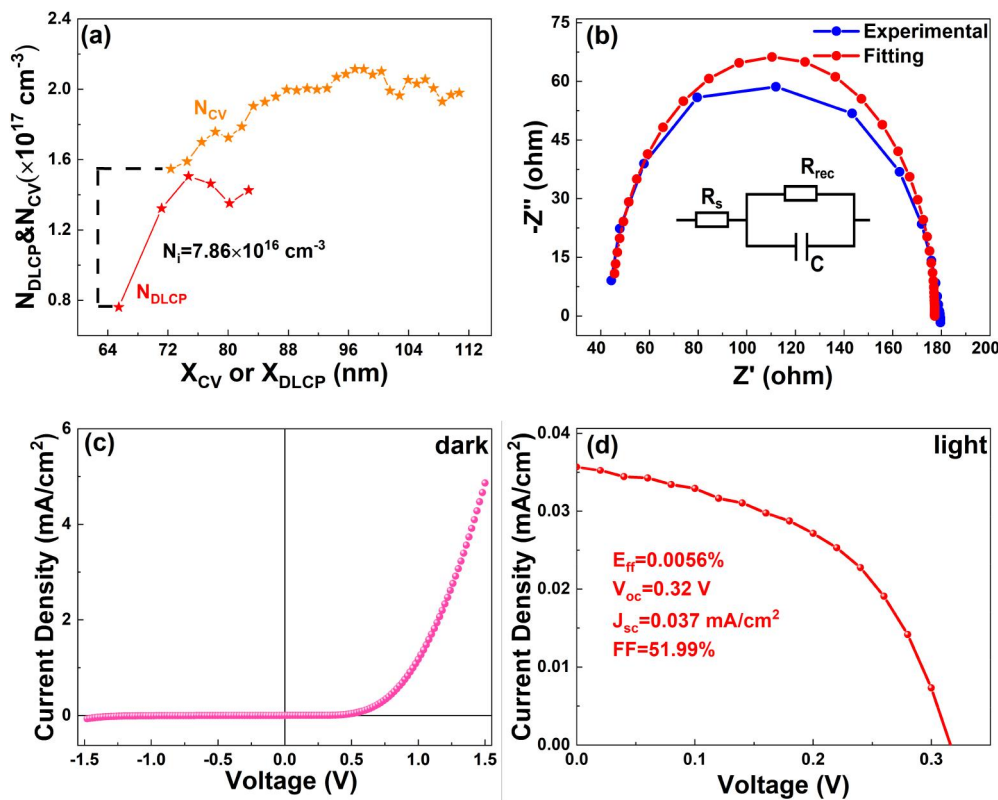


FIG. 5. (a) Extracting the interface state density. (b) The Nyquist plot (inset: the equivalent circuit for fitting). The dark (a) and illuminated (b) J-V curve of the homojunction with thinner ZTN and thicker AZTN.

cells,^{19,20} and implying higher minority carrier recombination rate, mainly due to the lower R_{rec} , which is another factor⁵⁹ leading to the relatively low E_{ff} of the present homojunction.

By reducing the deposition time of ZTN and increasing that of AZTN (detailed in [supplementary material](#)), the photovoltaic performance can be improved [Figs. 5(c) and 5(d)], demonstrating the promise for improving the performance.

In conclusion, the properties of AZTN/ZTN homojunction solar cells prepared by magnetron sputtering are studied, and the mechanism that limits the photovoltaic performance is revealed. The abrupt homojunctions exhibit pronounced rectification under dark conditions and a relatively weak photovoltaic response under illumination. The current transport is dominated by TE in 190–320 K and GR in the space charge region at higher temperature. Its J_{sc} is limited mainly by the high effective defect density and IR. In addition to IR, its V_{oc} is also affected by barrier height inhomogeneity or the temperature fluctuation of interface barrier height for holes. A shallow acceptor level at $E_V + 0.13 \text{ eV}$ and a deep interface state level at $E_V + 0.72 \text{ eV}$ are observed. The relatively high interface state density and low recombination resistance also limit the E_{ff} . The promise for improving the photovoltaic performance of the homojunctions is demonstrated.

See the [supplementary material](#) for details on the following: preparing and characterizing the films, including ZTN, AZTN, and Pt and the homojunctions; the steps for preparing the homojunctions,

XRD patterns, and SEM images; the reflectance, transmittance, and Tauc plots; the energy band diagram; analyzing the current transport mechanism; analyzing the barrier height inhomogeneity with the Gaussian distribution model; extracting the parameters in the single-diode model; the parameters extracted with the single-diode model and the calculated barrier heights; the curve of $\ln J_0$ vs $1/(kT)$; obtaining the standard deviation σ for ϕ_b ; fitting the forward-bias currents in 330–380 K; the detail about the new homojunctions in which the deposition time of ZTN is 40 min and that of AZTN is 4 h.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Fan Ye: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal);

Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Qian Gao**: Data curation (equal); Formal analysis (equal); Investigation (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Qiao Tang**: Data curation (supporting); Investigation (supporting). **Yi-Zhu Xie**: Resources (equal). **Ying Liu**: Resources (equal). **Chun-Yan Yu**: Resources (equal). **Fan Wang**: Resources (equal). **Dong-Ping Zhang**: Resources (equal). **Xing-Min Cai**: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- N. S. Lewis, "Research opportunities to advance solar energy utilization," *Science* **351**(6271), aad1920 (2016).
- T. R. Paudel and W. R. L. Lambrecht, "First-principles study of phonons and related ground-state properties and spectra in Zn-IV-N₂ compounds," *Phys. Rev. B* **78**(11), 115204 (2008).
- N. Kim, J. W. Choi, W. Song, S. Kang, S. Yim, J. W. Yoo, T. M. Chung, and K. S. An, "Exploring p-type conductivity and homojunction devices in amorphous zinc tin nitride: A step toward enhanced nitride semiconductor applications," *J. Alloys Compd.* **1040**, 183625 (2025).
- I. Gharibshahian and A. A. Orouji, "Interface engineering of cadmium-free Sn_{1-x}Ge_xO buffer layer for enhanced open-circuit voltage in Sb₂Se₃ solar cells," *Sol. Energy* **309**, 114466 (2026).
- A. Wang, J. L. Cong, S. J. Zhou, J. L. Huang, J. W. Cao, X. Cui, X. J. Yuan, Y. Yao, Z. Xu, G. J. He, J. Z. Liu, J. M. Cairney, Y. S. Chen, M. A. Green, S. H. Wei, K. W. Sun, and X. J. Hao, "Hydrogen-enhanced carrier collection enabling wide-bandgap Cd-free Cu₂ZnSnS₄ solar cells with 11.4% certified efficiency," *Nat. Energy* **10**(2), 255–265 (2025).
- S. Y. Chen, P. Narang, H. A. Atwater, and L. W. Wang, "Phase stability and defect physics of a ternary ZnSnN₂ semiconductor: First principles insights," *Adv. Mater.* **26**(2), 311–315 (2014).
- L. Lahourcade, N. C. Coronel, K. T. Delaney, S. K. Shukla, N. A. Spaldin, and H. A. Atwater, "Structural and optoelectronic characterization of RF sputtered ZnSnN₂," *Adv. Mater.* **25**(18), 2562–2566 (2013).
- J. Pan, J. Cordell, C. J. Tucker, A. C. Tamboli, A. Zakutayev, and S. Lany, "Interplay between composition, electronic structure, disorder, and doping due to dual sublattice mixing in nonequilibrium synthesis of ZnSnN₂O," *Adv. Mater.* **31**(11), 1807460 (2019).
- P. C. Quayle, K. L. He, J. Shan, and K. Kash, "Synthesis, lattice structure, and band gap of ZnSnN₂," *MRS Commun.* **3**(3), 135–138 (2013).
- A. Skrylev, A. Nezhdanov, I. Belokon, D. Usanov, A. Kudrin, D. Shestakov, L. Vinogradova, M. Baratta, G. De Filipo, and A. Mashin, "Electrophysical properties of cation-disordered ZnSnN₂ films: Temperature dependences and thermal annealing," *Physica B* **721**, 418058 (2026).
- A. Almushawwah, H. Albrithen, J. Labis, M. Al-Gawati, A. M. El-naggar, F. Alnjiman, J. Ghanbaja, J. F. Pierson, and A. N. Alodhayb, "Investigating Mg-doping on ZnSnN₂ ternary thin films via reactive magnetron co-sputtering," *Mater. Today Commun.* **51**, 114671 (2026).
- J. Y. Ren, L. Y. Liang, X. H. Liu, and H. T. Cao, "Physical properties of Zn-Sn-N films governed by the Zn/(Zn plus Sn) ratio," *J. Vac. Sci. Technol., A* **41**(3), 033406 (2023).
- X. J. Wu, F. Z. Meng, D. L. Chug, M. C. Yao, K. Guan, D. D. Zhang, and J. Meng, "Carrier tuning in ZnSnN₂ by forming amorphous and microcrystalline phases," *Inorg. Chem.* **58**(13), 8480–8485 (2019).
- A. N. Fioretti, A. Stokes, M. R. Young, B. Gorman, E. S. Toberer, A. C. Tamboli, and A. Zakutayev, "Effects of hydrogen on acceptor activation in ternary nitride semiconductors," *Adv. Electron. Mater.* **3**(3), 1600544 (2017).
- J. W. Choi, J. Kim, S. R. Pae, J. Kim, C. G. Kim, H. Lee, S. Ji, S. W. Nam, Y. K. Lee, and B. Shin, "Oxidation-resistant amorphous zinc tin nitride films with tunable optical and electrical properties," *Chem. Mater.* **34**(15), 6802–6808 (2022).
- F. Ye, Z. Ke, L. H. Yuan, R. T. Hong, D. P. Zhang, P. Fan, and X. M. Cai, "Electrical and optical properties of nanocrystalline ZnSnN₂," *Thin Solid Films* **772**, 139804 (2023).
- K. Javaid, W. H. Wu, J. Wang, J. F. Fang, H. L. Zhang, J. H. Gao, Z. G. Fei, L. Y. Liang, and H. T. Cao, "Band offset engineering in ZnSnN₂-based heterojunction for low-cost solar cells," *ACS Photonics* **5**(6), 2094–2099 (2018).
- F. Ye, R. T. Hong, C. S. He, Z. C. Zhao, Y. Z. Xie, D. P. Zhang, F. Wang, J. W. Li, and X. M. Cai, "ZnSnN₂ Schottky barrier solar cells," *Mater. Sci. Eng., B* **300**, 117097 (2024).
- F. Ye, C. S. He, T. Wu, S. Chen, Z. H. Su, X. H. Zhang, X. M. Cai, and G. X. Liang, "Sputtering deposited and energy band matched ZnSnN₂ buffer layers for highly efficient Cd-free Cu₂ZnSnS₄ solar cells," *Adv. Funct. Mater.* **34**(38), 2402762 (2024).
- F. Ye, Z. C. Zhao, C. S. He, J. L. Liang, Q. Gao, Y. Z. Xie, D. P. Zhang, and X. M. Cai, "Capacitance characterization and current transport mechanism of ZnSnN₂ heterojunctions," *Appl. Phys. Lett.* **125**(24), 242102 (2024).
- F. Ye, Q. Gao, Z. C. Zhao, J. L. Liang, Y. Z. Xie, D. P. Zhang, and X. M. Cai, "The barrier height formation mechanism and bulk inhomogeneity in ZnSnN₂/Cu Schottky junctions," *Appl. Phys. Lett.* **127**(10), 102101 (2025).
- J. H. Kim, J. Hwang, S. J. Yoon, J. Kim, Y. K. Lee, K. Park, and H. E. Lee, "Monolithic green-sensitive photodetectors enabled by a ZnSnN₂/GaN nanorods/silicon double heterojunction," *Inorg. Chem. Front.* **12**(5), 1958–1968 (2025).
- K. K. Chinnakutti, L. Patra, V. Panneerselvam, D. Govindarajan, S. Kheawhom, J. Theerthagiri, Y. Yu, S. T. Salammal, and M. Y. Choi, "Lithium inserted ZnSnN₂ thin films for solar absorber: N to p-type conversion," *Mater. Today Chem.* **25**, 100957 (2022).
- K. K. Chinnakutti, V. Panneerselvam, and S. T. Salammal, "Ba-acceptor doping in ZnSnN₂ by reactive RF magnetron sputtering: (002) faceted Ba-ZnSnN₂ films," *J. Alloys Compd.* **855**, 157380 (2021).
- X. M. Cai, J. L. Liang, Q. Gao, A. Hassan, U. Younis, X. Q. Tian, D. P. Zhang, and F. Ye, "The p-type conduction and Schottky junction of Ag-doped ZnSnN₂," *J. Appl. Phys.* **139**(15), 155703 (2026).
- F. Ye, R. T. Hong, Y. B. Qiu, Y. Z. Xie, D. P. Zhang, P. Fan, and X. M. Cai, "Nanocrystalline ZnSnN₂ prepared by reactive sputtering, its Schottky diodes and heterojunction solar cells," *Nanomaterials* **13**(1), 178 (2022).
- M. A. Rahman, S. M. N. Mamun, A. M. A. Hossain, and C. Ton-That, "ZnO nanorods on Li-doped ZnO thin films for efficient p-n homojunction light-emitting diodes," *ACS Appl. Nano Mater.* **6**(17), 15757–15763 (2023).
- Q. F. Dong, Y. J. Fang, Y. C. Shao, P. Mulligan, J. Qiu, L. Cao, and J. S. Huang, "Electron-hole diffusion lengths > 175 μm in solution-grown CH₃NH₃PbI₃ single crystals," *Science* **347**(6225), 967–970 (2015).
- P. K. Vasudev, B. L. Mattes, E. Pietras, and R. H. Bube, "Excess capacitance and non-ideal Schottky barriers on GaAs," *Solid-State Electron.* **19**(7), 557–559 (1976).
- J. G. Lu, Z. Z. Ye, G. D. Yuan, Y. J. Zeng, F. Zhuge, L. P. Zhu, B. H. Zhao, and S. B. Zhang, "Electrical characterization of ZnO-based homojunctions," *Appl. Phys. Lett.* **89**(5), 053501 (2006).
- L. Kosyachenko and T. Toyama, "Current-voltage characteristics and quantum efficiency spectra of efficient thin-film CdS/CdTe solar cells," *Sol. Energy Mater. Sol. Cells* **120**, 512–520 (2014).
- B. N. M. Ghadhab and A. Bahrami, "A comprehensive study on ZnSnN₂-based solar cells with different Cu-based delafossite as buffer layers: A numerical analysis using SCAPS-1D," *Sol. Energy Mater. Sol. Cells* **289**, 113675 (2025).
- S. Chand and J. Kumar, "Current-voltage characteristics and barrier parameters of Pd₂Si/p-Si(111) Schottky diodes in a wide temperature range," *Semicond. Sci. Technol.* **10**(12), 1680–1688 (1995).
- T. F. Schulze, L. Korte, E. Conrad, M. Schmidt, and B. Rech, "Electrical transport mechanisms in a-Si:H/c-Si heterojunction solar cells," *J. Appl. Phys.* **107**(2), 023711 (2010).

- ³⁵S. K. Cheung and N. W. Cheung, "Extraction of schottky diode parameters from forward current-voltage characteristics," *Appl. Phys. Lett.* **49**(2), 85–87 (1986).
- ³⁶J. H. Werner and H. H. Guttler, "Barrier inhomogeneities at Schottky contacts," *J. Appl. Phys.* **69**(3), 1522–1533 (1991).
- ³⁷J. H. Werner and H. H. Guttler, "Transport properties of inhomogeneous Schottky contacts," *Phys. Scr.* **T39**, 258–264 (1991).
- ³⁸M. Isik, O. Surucu, and N. M. Gasanly, "Temperature-dependent current voltage characteristics of p-GaSe_{0.75}Si_{0.25}/n-Si heterojunction," *Appl. Phys. A* **129**(8), 538 (2023).
- ³⁹S. S. Hegedus and W. N. Shafarman, "Thin-film solar cells: Device measurements and analysis," *Prog. Photovoltaics* **12**(2–3), 155–176 (2004).
- ⁴⁰C. R. Crowell and V. L. Rideout, "Normalized thermionic-field (T-F) emission in metal-semiconductor (Schottky) Barriers," *Solid-State Electron.* **12**(2), 89–105 (1969).
- ⁴¹S. C. Lee and G. L. Pearson, "Current-voltage characteristics of Al_xGa_{1-x}As Schottky barrier and p-n junctions," *J. Appl. Phys.* **52**(8), 5202–5206 (1981).
- ⁴²U. Rau and H. W. Schock, "Electronic properties of Cu(In,Ga)Se₂ heterojunction solar cells-recent achievements, current understanding, and future challenges," *Appl. Phys. A* **69**(2), 131–147 (1999).
- ⁴³C. Frisk, T. Ericson, S. Y. Li, P. Szaniawski, J. Olsson, and C. Platzer-Björkman, "Combining strong interface recombination with bandgap narrowing and short diffusion length in Cu₂ZnSnS₄ device modeling," *Sol. Energy Mater. Sol. Cells* **144**, 364–370 (2016).
- ⁴⁴U. Rau and J. H. Werner, "Radiative efficiency limits of solar cells with lateral band-gap fluctuations," *Appl. Phys. Lett.* **84**(19), 3735–3737 (2004).
- ⁴⁵V. Nadenau, U. Rau, A. Jasenek, and H. W. Schock, "Electronic properties of CuGaSe₂-based heterojunction solar cells.: Part I.: Transport analysis," *J. Appl. Phys.* **87**(1), 584–593 (2000).
- ⁴⁶C. J. Hages, N. J. Carter, R. Agrawal, and T. Unold, "Generalized current-voltage analysis and efficiency limitations in non-ideal solar cells: Case of Cu₂ZnSn(S_xSe_{1-x})₄ and Cu₂Zn(Sn_yGe_{1-y})(S_xSe_{1-x})₄," *J. Appl. Phys.* **115**(23), 234504 (2014).
- ⁴⁷X. M. Cai, B. Wang, F. Ye, K. Vaithinathan, J. J. Zeng, D. P. Zhang, P. Fan, and V. A. L. Roy, "Tuning the photoluminescence, conduction mechanism and scattering mechanism of ZnSnN₂," *J. Alloys Compd.* **779**, 237–243 (2019).
- ⁴⁸M. Pandey, K. Kuhar, and K. W. Jacobsen, "II-IV-V₂ and III-III-V₂ polytypes as light absorbers for single junction and tandem photovoltaic devices," *J. Phys. Chem. C* **121**(33), 17780–17786 (2017).
- ⁴⁹A. Punya, W. R. L. Lambrecht, and M. van Schilfgaarde, "Quasiparticle band structure of Zn-IV-N₂ compounds," *Phys. Rev. B* **84**(16), 165204 (2011).
- ⁵⁰E. Arca, A. Fioretti, S. Lany, A. C. Tamboli, G. Teeter, C. Melamed, J. Pan, K. N. Wood, E. Toberer, and A. Zakutayev, "Band edge positions and their impact on the simulated device performance of ZnSnN₂-based solar cells," *IEEE J. Photovoltaics* **8**(1), 110–117 (2018).
- ⁵¹L. A. Kosyachenko, O. L. Maslyanchuk, V. V. Motushchuk, and V. M. Sklyarchuk, "Charge transport generation-recombination mechanism in Au/n-CdZnTe diodes," *Sol. Energy Mater. Sol. Cells* **82**(1–2), 65–73 (2004).
- ⁵²G. Brammertz, S. Oueslati, M. Buffière, J. Bekaert, H. El Anzeery, K. Ben Messaoud, S. Sahayaraj, T. Nuytten, C. Köble, M. Meuris, and J. Poortmans, "Investigation of properties limiting efficiency in Cu₂ZnSnSe₄-based solar cells," *IEEE J. Photovoltaics* **5**(2), 649–655 (2015).
- ⁵³M. M. El-Nahass, H. M. Zeyada, M. S. Aziz, and N. A. El-Ghamaz, "Carrier transport mechanisms and photovoltaic properties of Au/p-ZnPC/p-Si solar cell," *Solid-State Electron.* **49**(8), 1314–1319 (2005).
- ⁵⁴H. Elangovan, A. V. Kesavan, K. Chattopadhyay, and P. C. Ramamurthy, "2D layering of silicon nanocrystals at TiO₂/CuI heterojunction for enhanced charge transport," *J. Appl. Phys.* **125**(24), 245302 (2019).
- ⁵⁵I. Hussain, M. Y. Soomro, N. Bano, O. Nur, and M. Willander, "Interface trap characterization and electrical properties of Au-ZnO nanorod Schottky diodes by conductance and capacitance methods," *J. Appl. Phys.* **112**(6), 064506 (2012).
- ⁵⁶C. Barret and A. Vapaille, "Interfacial states spectrum of a metal-silicon junction," *Solid-State Electron.* **19**(1), 73–75 (1976).
- ⁵⁷A. N. Fioretti, J. Pan, B. R. Ortiz, C. L. Melamed, P. C. Dippo, L. T. Schelhas, J. D. Perkins, D. Kuciauskas, S. Lany, A. Zakutayev, E. S. Toberer, and A. C. Tamboli, "Exciton photoluminescence and benign defect complex formation in zinc tin nitride," *Mater. Horiz.* **5**(5), 823–830 (2018).
- ⁵⁸C. E. Michelson, A. V. Gelatos, and J. D. Cohen, "Drive-level capacitance profiling: Its application to determining gap state densities in hydrogenated amorphous silicon films," *Appl. Phys. Lett.* **47**(4), 412–414 (1985).
- ⁵⁹C. H. Jiang, R. F. Tang, X. M. Wang, H. X. Ju, G. L. Chen, and T. Chen, "Alkali metals doping for high-performance planar heterojunction Sb₂S₃ solar cells," *Sol. RRL* **3**(1), 1800272 (2019).